program 9

AUTHOR
kumkummk 1nt23is106

9. Create multiple histograms using ggplot2 :: facet_wrap() to visualize how a variable (e.g sepal.length) is distributed across different groups (e.g species) in a built_in R dataset

Step1: Load required library

```
library(ggplot2)
```

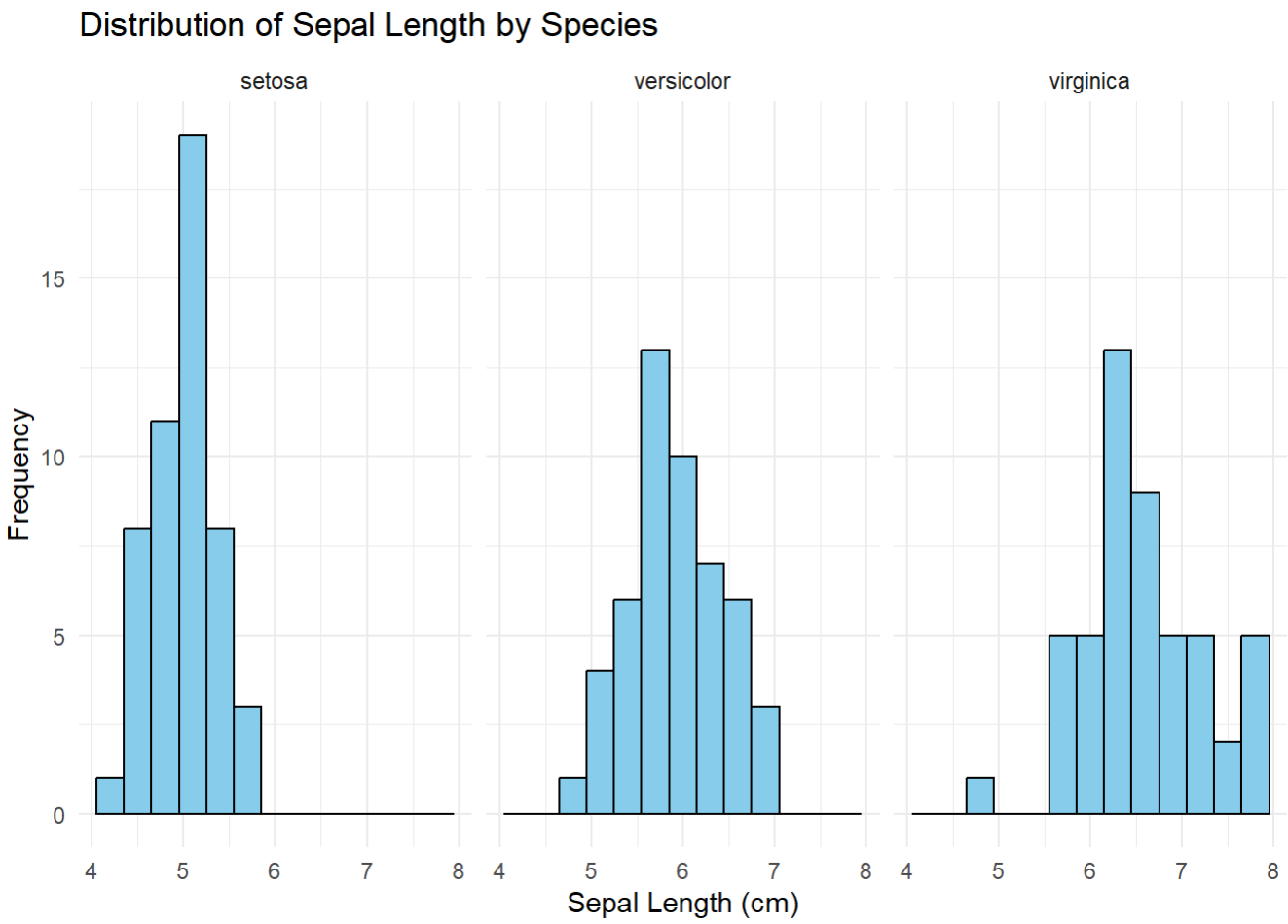
Step 2: Load Built-in Dataset

```
data <- iris
head(data)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
```

Step 3: Create histogram of Sepal.length grouped by Species using facet_wrap

```
# Create histogram of Sepal.Length grouped by Species using facet_wrap
ggplot(data = iris, aes(x = Sepal.Length)) +
  geom_histogram(binwidth = 0.3, fill = "skyblue", color = "black") +
  facet_wrap(~ Species) +
  labs(
    title = "Distribution of Sepal Length by Species",
    x = "Sepal Length (cm)",
    y = "Frequency"
  ) +
  theme_minimal()
```


Distribution of Sepal Length by Species
setosa
versicolor
virginica
Frequency
0
5
10
15
4
5
6
7
8
Sepal Length (cm)